Vedaksha: A Clean-Room, Sub-Arcsecond Ephemeris Engine for Computational Vedic Astrology

Amit Gupta
ArthIQ Labs LLC
info@arthiq.net
ORCID: 0009-0008-9450-8134

2026-10-24

Technical report describing Vedaksha v10.0.0. Not peer reviewed.
Measurements and explicit gaps are version-specific.

Abstract

Vedaksha is a clean-room ephemeris and Vedic astrology (jyotish) computation engine written in Rust. It computes planetary positions from the analytical VSOP87A and ELP/MPP02 series and from the JPL DE440s numerical kernel. Re-measured on 2026-10-24 against NASA/JPL Horizons DE441, the numerical path has a mean apparent geocentric ecliptic-longitude residual of 0.096 arcseconds and a maximum of 0.272 arcseconds over 15,350 comparisons spanning 1900–2025. The analytical path has a mean of 0.160 arcseconds and a maximum of 1.473 arcseconds over 13,815 comparisons in the same interval. The sub-arcsecond description refers to mean longitude agreement, not a bound on every analytical result or on other coordinates. From 1962-01-20, the oracle converts Horizons' UTC time tags to Terrestrial Time using the historical TAI minus UTC record, removing both ΔT and UT1 minus UTC from that part of the comparison. Version 10 corrects the nutation coefficients and the VSOP87A frame rotation, includes solar gravitational light deflection, and updates ΔT from measured and predicted records. The engine exposes derived jyotish computations through native Rust, Python, WebAssembly, and Model Context Protocol interfaces. Primary sources and derivation records are documented in a public provenance ledger. An external parity harness measures the analytical path against JPL DE440 and IMCCE INPOP21a and over six centuries either side of J2000.0. The software is source-available under BUSL-1.1, with conversion to Apache-2.0 four years after each version's release.

1 Introduction

An ephemeris result is defined by more than a planetary theory: its time scale, target, reference frame, and apparent-place corrections determine what a longitude means. Vedaksha makes these choices explicit and records the sources of its algorithms and data. ArthIQ Labs, an AI platforms company, develops the engine for computational jyotish and machine-accessible astronomical calculations.

Here "clean-room" describes a source and derivation procedure. The analytical theories come from their authors' publications and coefficient distributions; the numerical kernel comes from NASA/JPL. The repository's `DATA_PROVENANCE.md` records source URLs, retrieval dates, licensing, and hashes where established. Dated records under `docs/audit/` document the lunar-theory and

ayanamsha derivations, including the replacement of early sidereal-offset constants by primary-source derivations. Those records support a procedural provenance claim; numerical agreement alone cannot establish one.

This report treats the ephemeris and sidereal-conversion methods in detail. The derived jyotish layer, including nakshatras, dashas, vargas, panchanga, shadbala, and muhurta, is described at the interface level. Its outputs do not inherit an accuracy guarantee merely because their input longitudes have been tested.

2 Methods

2.1 Analytical ephemeris pipeline

VSOP87A supplies planetary positions from Mercury through Neptune [Bretagnon and Francou, 1988]. ELP/MPP02 supplies lunar positions [Chapront and Francou, 2003].

The coefficient tree contains 331 files at this checkout, counted in the ephemeris crate's `src/analytical/coefficients/` directory. Generated binary tables are embedded in the program, so this path requires no runtime data file. Lunar phases are evaluated by angle addition through a shared prefix tree; VSOP87A reuses sine and cosine evaluations across terms with the same frequency. No performance benchmark was re-run for this report.

VSOP87A vectors are transformed from the dynamical ecliptic and equinox of J2000 to FK5 J2000 by the matrix published with the series in Bretagnon and Francou's `vsop87.doc`, "REFERENCE SYSTEM" [Bretagnon and Francou, 1988]. The implementation's `VSOP87A_TO_EQUATORIAL` applies this matrix to both position and velocity. Its elements were checked against the authors' distribution for this revision. This replaces the former rotation by the IAU 1976 obliquity alone, which omitted the published equinox rotation. The matrix targets FK5; the implementation applies no additional FK5-to-ICRS transformation. The Moon still uses the IAU 1976 J2000 obliquity, 84381.448 arcseconds. Its frame relation was not independently re-derived for this report, so the former claim that this rotation was verified for both theories is withdrawn.

2.2 Numerical ephemeris pipeline

The second implementation behind `EphemerisProvider` reads the DE440s SPK kernel from NAIF [NAIF/JPL, 2021, Park et al., 2021]. Vedaksha's DAF/SPK parser evaluates type-2 Chebyshev position segments and rejects requests outside the kernel's segment coverage. The kernel used here was downloaded and checked against the repository's pinned SHA256 before measurement. The comparison targets planetary-system barycentres, matching the segments DE440s stores, with the Sun and Moon addressed separately. A comparison against physical planet centres would measure a different quantity.

Horizons is an external validation service, not a kernel ingested by Vedaksha. Its DE441 reference is distinct from DE440s, but these are not independent observational lineages: DE440 and DE441 were fitted to the same data, with a different lunar-core damping assumption [Park et al., 2021]. Agreement tests the implementation against an external result without proving independence of all underlying models or observations.

2.3 Apparent place and reference frames

The apparent-place pipeline applies light-time and aberration corrections and, for planets, solar gravitational light deflection. The latter uses the finite-distance weak-field expression of Klioner

[2003, eqs. 75 and 77]. `deflection::solar_deflection` operates on the astrometric source vector and the observer's displacement from the Sun. The Sun and Moon are excluded. For a straight ray crossing the solar disc, the implementation returns zero displacement as an occultation convention; this is not a physical prediction of an observable undeflected ray. Deflection by other bodies and solar multipole terms are not modelled.

The frame of date uses IAU 2006 precession in the four-angle Fukushima and Williams parameterization [Capitaine et al., 2003], composed as $R_x(-\epsilon_A)R_z(-\bar{\psi})R_x(\bar{\phi})R_z(\bar{\gamma})$. Nutation is then applied before rotation to the ecliptic of date. The nutation table contains the first 77 luni-solar rows of IAU 2000A, the truncated series used for IAU 2000B [McCarthy and Luzum, 2003]. A row-by-row comparison with the IAU 2000A coefficients represented in ERFA confirms all 77 current rows. Relative to the preceding implementation, 30 rows required correction; 17 had the main longitude coefficient's sign reversed. Thus the previous report's unqualified description of the old table as the published series was incorrect. The implementation omits the full planetary terms, the fixed planetary-bias offsets of 2000B, and free core nutation. It should therefore be read as the corrected truncated luni-solar implementation, not as a claim to implement every part of the full IAU model.

2.4 Time scales and ΔT

The ordinary position API accepts UT1 and converts it to TT using $\Delta T = \mathrm{TT} - \mathrm{UT1}$. The TT-addressed `ecliptic_position_tt` entry point bypasses that conversion. At v10.0.0, `delta_t.rs` uses the following sources.

Interval	Source and evaluation
Before year −720	Espenak and Meeus long-term parabola.
−720 to 1962.0	Stephenson, Morrison and Hohenkerk cubic spline, Table S15.
1962.0 to 1973.0	USNO half-year measurements, linearly interpolated.
1973-02 to 2026-10	USNO monthly measurements, linearly interpolated.
2026-10 to 2033.75	Interpolation onto and through USNO's published predictions.
After 2033.75	Espenak and Meeus expressions shifted to join the last USNO prediction.

Table 1: The v10.0.0 ΔT source selection, using astronomical years. The measured-to-predicted handover is not the 2026-01-01 reporting cut used below.

The historical spline is from Stephenson et al. [2016]; the extrapolating expressions are from Espenak and Meeus [2006]. The USNO files are `historic_deltat.data`, `deltat.data`, and `deltat.preds`, with retrieval dates and hashes recorded in `DATA_PROVENANCE.md`. Evaluation uses a continuous calendar decimal year rather than a constant value within each month. The source joins at −720 and 1962.0 retain documented discontinuities; the remaining joins are continuous. The shifted tail is an extrapolation, not a measurement of future Earth rotation.

2.5 Sidereal (ayanamsha) conversion

Sidereal longitude subtracts the true ayanamsha from apparent tropical longitude. The true ayanamsha is the mean precession-only offset plus nutation in longitude; using the mean offset alone would retain a nutation term in the sidereal result. Both forms have explicitly named APIs.

The engine implements eleven sidereal systems, in addition to a tropical option. The ayanamsha audit dated 2026-08-17 records their normative definitions and declared assumptions in `spec.md`

and `derivation-inputs.json`. These include the Indian official ayanamsha from *The Indian Astronomical Ephemeris*, the Fagan and Bradley, Krishnamurti, and Raman book anchors, the *Surya Siddhanta* (Ch. 3, vv. 9–12), Yukteshwar, and the Revati, True Chitra, Pushya, True Mula, and Galactic Centre systems. Star anchors use the Hipparcos re-reduction [van Leeuwen, 2007]; the Galactic Centre anchor uses the VLBI determination of Sagittarius A* [Gordon et al., 2023]. The Rust implementation is checked against a separately written Python derivation from the same input manifest. These share primary inputs and authorship, so agreement does not exclude a shared transcription error. The measured fixture result is reported below.

2.6 Provenance and verification boundary

The committed source ledger and dated derivation records identify the inputs consulted. For this revision, the nutation rows and VSOP87A matrix were checked against their reference distributions, and the DE440s and committed Horizons fixture hashes were verified. The lunar and planetary coefficient distributions were not regenerated, the historical clean-room procedures were not repeated, and the Horizons fixture was not fetched afresh. The longitude results below are new evaluations against the existing pinned fixture.

3 Validation

3.1 Oracle construction and time-tag matching

The fixture `tests/oracle_jpl/reference_positions.json` contains 24,350 rows: ten targets at 2,435 dates on a 30-day grid in 1900–2100, originally fetched on 2026-07-16 from Horizons [Giorgini, 2015]. The targets are the Sun, Moon, and planetary-system barycentres from Mercury through Pluto. The tested quantity is the absolute shortest-arc difference in apparent geocentric ecliptic longitude of date. The analytical provider has no Pluto model, so its oracle evaluates 21,915 rows.

Horizons' "UT" tag means UT1 before 1962-01-20 and UTC from that date. For the earlier rows, both tests call the UT1-addressed engine API, so the engine's ΔT remains in the residual. For later rows they use

$$\mathrm{JD_{TT}} = \mathrm{JD_{UTC}} + \frac{32.184 + (\mathrm{TAI} - \mathrm{UTC})}{86400},$$

with offsets in seconds, then call `ecliptic_position_tt`. The USNO historical TAI minus UTC table includes the pre-1972 rate terms as well as subsequent leap seconds. The last known correction is retained for future dates, matching Horizons' convention. This removes both ΔT and UT1 minus UTC from the comparison from 1962-01-20 onward. It does not validate the engine's future UT1-to-TT predictions or predict future leap seconds.

The 1900–2025 reporting window is retained to preserve the published metric's definition and comparison count. It is now a mixture of two time-tag regimes, not a boundary separating all measured time conversions from all predicted ones. Horizons also uses an older precession/nutation convention for its ecliptic of date than Vedaksha's IAU 2006/2000B pipeline. The residuals include this convention difference; they are not isolated measurements of theory truncation.

3.2 Re-measured longitude residuals

Both release-mode oracle tests were run on 2026-10-24 with fixtures required, so a missing kernel or reference file could not silently skip validation. The commands and logs are indexed in `docs/paper/README.md`. The ephemeris source, manifest, and lockfile used by these tests match tag v10.0.0.

Interval	n	Mean (arcsec)	Max (arcsec)
1900–2025	15,350	0.096	0.272 (Moon, 1962-01-06)
2026–2100	9,000	0.075	0.189 (Moon, 2099-10-04)
Whole fixture	24,350	0.088	0.272 (Moon, 1962-01-06)

Table 2: SPK longitude residuals against Horizons, re-measured 2026-10-24. No rows were skipped. The future-interval mean is aggregated from the same run's printed per-row residuals; the other means are printed by the test.

Body	SPK mean/max	Analytical mean/max
Sun	0.095/0.244	0.094/0.260
Moon	0.092/0.272	0.094/0.276
Mercury	0.095/0.245	0.094/0.278
Venus	0.095/0.243	0.094/0.269
Mars	0.095/0.244	0.090/0.246
Jupiter	0.096/0.246	0.148/0.335
Saturn	0.096/0.246	0.159/0.338
Uranus	0.095/0.246	0.257/1.201
Neptune	0.096/0.247	0.410/1.473
Pluto	0.100/0.251	n/a

Table 3: Longitude residuals in arcseconds over 1900–2025, with 1,535 comparisons per available body. Analytical Pluto is unavailable. Analytical totals are 13,815 comparisons, 0.160 mean and 1.473 maximum (Neptune, 2025-09-02). Across the whole analytical fixture, 21,915 comparisons give 0.251 mean and 2.248 maximum (Neptune, 2078-01-27).

Body	SPK before	SPK from	Analytical before	Analytical from
Sun	0.144/0.244	0.048/0.049	0.151/0.260	0.039/0.050
Moon	0.138/0.272	0.047/0.053	0.135/0.276	0.055/0.066
Mercury	0.144/0.245	0.048/0.051	0.151/0.278	0.039/0.053
Venus	0.144/0.243	0.048/0.051	0.150/0.269	0.038/0.066
Mars	0.144/0.244	0.048/0.051	0.145/0.246	0.038/0.132
Jupiter	0.145/0.246	0.048/0.051	0.131/0.227	0.165/0.335
Saturn	0.145/0.246	0.048/0.052	0.153/0.271	0.165/0.338
Uranus	0.144/0.246	0.048/0.050	0.147/0.293	0.365/1.201
Neptune	0.145/0.247	0.048/0.050	0.153/0.188	0.660/1.473
Pluto	0.146/0.251	0.054/0.062	n/a	n/a

Table 4: The 1900–2025 sample split at 1962-01-20. Each cell is mean/max longitude residual in arcseconds. The before-boundary columns have 756 comparisons per body and include the engine's ΔT; the from-boundary columns have 779 per body through 2025 and use UTC tags converted to TT.

The 2026–2100 SPK maximum is 0.189 arcseconds, for the Moon on 2099-10-04. These rows compare the same TT instants on both sides, so no future ΔT prediction enters them. The whole-fixture mean is measured directly from the current run. Where a mean is aggregated from the printed per-row residuals, its precision is limited to their three decimal places; the independently printed test summary is also reported. The rounded interval means reproduce the printed whole-fixture mean:

$$\frac{15{,}350 \times 0.096 + 9{,}000 \times 0.075}{24{,}350} \approx 0.088 \quad \text{arcseconds.}$$

The split at 1962-01-20 exposes the change in comparison semantics. Before the boundary, the residual includes the engine's UT1-to-TT conversion. After it, the residual measures the ephemeris and apparent-place conventions with the UTC-to-TT conversion supplied by the oracle. The largest 1900–2025 SPK residual falls on 1962-01-06, between the engine's 1962.0 source handover and Horizons' 1962-01-20 boundary. This is consistent with the source-handover difference documented in `MAINTENANCE.md` §10; no separate causal experiment was run here.

3.3 Cross-validation against a second agency

The external harness at `https://github.com/arthiqlabs/vedaksha-parity` compares published engine output against local JPL DE440 and IMCCE INPOP21a kernels. It is maintained by the same authors as Vedaksha. It is a separate test implementation, not an independent validation team. On 2026-10-24 it ran the published `vedaksha==10.0.0` Python package, which hosts the engine over WebAssembly with no SPK file loaded, at the 2,000 instants of its bundled birth-data sample (seed 42). It compared tropical apparent longitudes addressed in UT1 for seven bodies (Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn), reading both kernels through Skyfield 1.55; Mars, Jupiter and Saturn are answered from their system barycentres. The lunar nodes are in the case list, but neither kernel models them, so they are refused rather than compared. The counts differ because each adapter refuses instants outside its own validity window: for DE440 the ΔT window of roughly 1800–2100, for INPOP21a its kernel coverage. The configuration and run artifacts are committed in the harness under `configs/crossval-2000.yaml` and `results/canonical/`, and the figures are recorded in `docs/accuracy/2026-10-24-parity-results.json`.

Reference	Agency	n	Mean (arcsec)	Max (arcsec)
JPL DE440 (local kernel)	NASA/JPL	13,832	0.030	0.695 (Mars)
IMCCE INPOP21a (local kernel)	IMCCE	11,788	0.025	0.696 (Mars)

Table 5: Analytical-path cross-validation of v10.0.0 against two national-agency ephemerides, seven bodies at 2,000 sample instants. n counts body–instant comparisons.

Raw SPK state-vector comparisons and mean-node drift were not measured for this release, and no claim is made about either.

3.4 Architecture and reproducibility

The v10 lunar series builds phases by angle addition using scalar arithmetic and pure-Rust libm; it no longer uses the SIMD trigonometric kernel. The committed lunar-series digest pins are shared between aarch64 and x86-64. VSOP87A still uses the SIMD kernel, and the combined analytical-output digest remains architecture-specific. These are regression pins, not an accuracy measurement. Digest tests were not re-run for this report.

Architecture	n	Mean (arcsec)	Max (arcsec)
aarch64 (native macOS)	13,815	0.160	1.473 (Neptune, 2025-09-02)
x86-64 (macOS, Rosetta)	13,815	0.160	1.473 (Neptune, 2025-09-02)

Table 6: Analytical-path residuals over 1900–2025, re-measured at this checkout with Rust 1.99.1. The x86-64 binary used the default target configuration under Rosetta, not a native x86-64 machine or an explicitly selected AVX2 build. Every printed per-body mean and maximum agrees at three decimal places.

Agreement to the printed precision does not establish bit identity of all outputs, or cover hardware and compiler configurations that were not measured.

3.5 Full-range accuracy

The same harness sampled the analytical path against DE440 at 43 instants spaced 9,131.25 days (25 Julian years) apart from JD(TT) 2292184.5, inside DE440's 1550–2650 coverage, through the TT-addressed entry point. Both sides are addressed in TT, so no ΔT model enters either side; the reference is Skyfield's apparent place of date against the same kernel. Table 7 groups the absolute longitude residuals by distance from J2000.0. Sample counts fall in the outer bands as the kernel's coverage margin approaches, and the one instant beyond 600 years falls in no band. The sweep is `scripts/fullrange_tt_sweep.py` in the harness, and its output is committed there as `results/canonical/fullrange-tt-10.0.0-2026-10-24.json`.

Body	0–100	100–200	200–300	300–400	400–500	500–600
n	8	8	8	8	6	4
Sun	0.010/0.025	0.033/0.056	0.053/0.099	0.073/0.160	0.069/0.181	0.055/0.075
Moon	0.010/0.019	0.016/0.028	0.045/0.087	0.078/0.167	0.241/0.354	0.508/0.602
Mercury	0.012/0.053	0.031/0.069	0.056/0.160	0.072/0.140	0.108/0.266	0.068/0.105
Venus	0.009/0.025	0.034/0.050	0.054/0.094	0.071/0.136	0.120/0.273	0.019/0.040
Mars	0.027/0.077	0.143/0.393	0.268/0.735	0.443/0.849	1.347/3.881	2.394/6.371
Jupiter	0.141/0.292	0.304/0.661	0.442/0.516	0.412/0.741	0.450/0.782	0.443/0.505
Saturn	0.146/0.217	0.200/0.347	0.361/0.495	0.501/0.609	0.752/0.935	1.075/1.231
Uranus	0.544/1.312	0.611/2.183	1.365/3.043	1.849/4.499	3.117/6.493	5.395/10.056
Neptune	0.982/2.202	1.910/3.538	3.377/4.388	4.805/6.368	5.768/6.280	7.725/8.196

Table 7: Analytical longitude residuals of v10.0.0 against DE440, grouped by distance from J2000.0 in years, both sides addressed in TT. Each cell is mean/max of the absolute residual in arcseconds; n is the number of sampled instants in the band. With at most eight samples per cell, and a 25-year grid sampling any short-period term at an effectively arbitrary phase, each maximum is only the largest sampled residual, a lower bound on the worst case over its interval.

Residuals grow with distance from J2000.0 for every body, most for Mars and the outer planets. This is the precision of the VSOP87A theory itself, which its authors fitted to the DE200 integration (`vsop87.doc`, "PRECISION"): the engine evaluates the published series with the authors' published frame rotation and applies no empirical correction toward any modern ephemeris. For the outer planets over 1900–2100, the untruncated series gives the same residuals to four decimals (`MAINTENANCE.md` §10), so truncation is not the cause there; that check was not repeated at these epochs. Applications needing sub-arcsecond planetary positions centuries from J2000.0 should use the numerical SPK path with a kernel covering the interval. At the same 43 instants, the SPK

path loaded with the full DE440 kernel agrees with Skyfield's apparent place from that kernel to 0.003 arcseconds at worst, with no growth away from J2000.0. Because both sides read one kernel, this checks only the apparent-place pipeline, not the kernel's own accuracy. Addressing both sides in TT removes ΔT from this experiment but does not eliminate differences in apparent-place conventions.

3.6 Scope and other checks

The longitude oracle does not measure latitude, distance, speed, lunar-node longitude, house cusps, or derived jyotish events. A planetary-system barycentre is not a lunar node: the latter requires a geometric or secular definition in addition to an ephemeris. Node accuracy and secular drift were not measured for this release, so no sub-arcsecond node claim is made. The supported API range is also not a measured accuracy range; beyond the fixture dates, the only evidence is the sparse full-range sample in Table 7.

The ayanamsha fixture test made 192 comparisons against the committed Python-derived reference and measured a worst disagreement of 5.684×10^{-14} degrees (Revati-paksha at year 285 CE). Its separate coverage check also passed. This is agreement between two evaluations of shared primary inputs, not an external accuracy bound on the sidereal definitions. The three tests in `oracle_time_scale.rs` also passed, checking historical TAI minus UTC values, the time-tag boundary, and the conversion against committed Horizons time-offset samples.

The release's `metrics.json` records 1,603 per-push tests and 1,614 full-suite tests, measured at tag v10.0.0. These are tag measurements, not counts from this report's targeted runs. The full workspace suite was not re-run.

4 Software Architecture and Availability

The chart representation is a property graph. At this checkout, the graph crate's `ontology.rs` defines nine node types and twelve edge types. The `emit_graph` MCP tool exports Cypher, SurrealQL, JSON-LD, JSON, and embedding text. The committed `tools/mcp-tools.json` snapshot contains 17 tools, covering charts, dashas, karakas, combustion, shadbala, vargas, graph export, transits and searches, muhurta, ashtakavarga, gochara, panchanga, drishti, bhavas, synastry, and composite charts.

Surface	Interface or installation
Native Rust	`cargo add vedaksha`
Python	`pip install vedaksha`
WebAssembly	`npm install vedaksha-wasm`
MCP container	`ghcr.io/arthiqlabs/vedaksha-mcp`
Source	`https://github.com/arthiqlabs/vedaksha`

Table 8: Distribution interfaces described by the v10.0.0 checkout. Live registry versions, package installation, and container execution were not verified for this report.

Vedaksha is source-available under the Business Source License 1.1 (BUSL-1.1). The repository's `LICENSE` permits non-commercial use under its Additional Use Grant and requires a commercial license for the commercial uses it defines. The stated fee is USD $500 once per organization, covering unlimited products and seats perpetually and including subsequent versions. Each version

converts to Apache License 2.0 four years after its release. This report is not peer reviewed and has not been submitted for publication. No archival software DOI is claimed.

5 Conclusion

The v10.0.0 numerical path agrees with the pinned Horizons longitude reference to 0.096 arcseconds mean and 0.272 arcseconds maximum over 1900–2025; the analytical path measures 0.160 arcseconds mean and 1.473 arcseconds maximum. From 1962-01-20 the comparison uses the same TT instants on both sides, so ΔT does not enter it. These results exercise the corrected nutation table, published VSOP87A rotation, and solar-deflection path, while preserving explicit limits on what was measured. Against two national-agency ephemerides the analytical path measures 0.030 and 0.025 arcseconds mean, 0.696 arcseconds maximum, over seven bodies. Node and house-cusp accuracy claims await the corresponding evidence.

References

P. Bretagnon and G. Francou. Planetary theories in rectangular and spherical variables: VSOP87 solution. *Astronomy & Astrophysics*, 202:309, 1988.

N. Capitaine, P. T. Wallace, and J. Chapront. Expressions for IAU 2000 precession quantities. *Astronomy & Astrophysics*, 412:567–586, 2003.

J. Chapront and G. Francou. The lunar theory ELP revisited. Introduction of new planetary perturbations. *Astronomy & Astrophysics*, 404:735–742, 2003. doi: 10.1051/0004-6361:20030529.

Fred Espenak and Jean Meeus. Five millennium canon of solar eclipses: -1999 to +3000. Technical Report TP-2006-214141, NASA, 2006. URL `https://eclipse.gsfc.nasa.gov/SEpubs/5MCSE.html`.

J. D. Giorgini. Status of the JPL Horizons ephemeris system. In *IAU General Assembly*, volume 29, 2015.

D. Gordon, A. de Witt, and C. S. Jacobs. Position and proper motion of Sagittarius A* in the ICRF3 frame from VLBI absolute astrometry. *The Astronomical Journal*, 165:49, 2023. doi: 10.3847/1538-3881/aca65b.

Sergei A. Klioner. A practical relativistic model for microarcsecond astrometry in space. *The Astronomical Journal*, 125:1580–1597, 2003. doi: 10.1086/367593.

Dennis D. McCarthy and Brian J. Luzum. An abridged model of the precession-nutation of the celestial pole. *Celestial Mechanics and Dynamical Astronomy*, 85:37–49, 2003. doi: 10.1023/A:1021762727016.

NAIF/JPL. JPL DE440s SPK kernel (numerical planetary and lunar ephemeris). `https://naif.jpl.nasa.gov/pub/naif/generic_kernels/spk/planets/de440s.bsp`, 2021.

Ryan S. Park, William M. Folkner, James G. Williams, and Dale H. Boggs. The JPL planetary and lunar ephemerides DE440 and DE441. *The Astronomical Journal*, 161:105, 2021. doi: 10.3847/1538-3881/abd414.

F. R. Stephenson, L. V. Morrison, and C. Y. Hohenkerk. Measurement of the Earth's rotation: 720 BC to AD 2015. *Proceedings of the Royal Society A*, 472:20160404, 2016. doi: 10.1098/rspa.2016.0404.

F. van Leeuwen. Validation of the new Hipparcos reduction. *Astronomy & Astrophysics*, 474: 653–664, 2007.